

Expanding Polynomial Dynamics over Surreal and Surcomplex Fields

Exact symbolic fibers and the order-unit obstruction

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A single-source report of the collection, docs/surcomplex/expanding-polynomial-dynamics

Abstract

Let $K = k((t^\Gamma))$, with $k = \mathbb{R}$ or $\mathbb{C}$ and Γ an arbitrary nonzero set-sized ordered abelian group. We study $F = q^{-1}P$, where $v(q) = \kappa > 0$, $\deg P = \deg \text{red } P = d \geq 2$, and the reduction of $P \in O[X]$ splits into distinct roots in k . We construct, by finite-parameter formal algebra and strong Hahn evaluation, a canonical point x_s for every one-sided d -symbol itinerary. The entire integral nonescaping set is the disjoint union $\bigsqcup_s (x_s + I_\kappa)$, where $I_\kappa = \{h : v(h) > n\kappa \text{ for every } n \in \mathbb{N}\}$. Different itineraries first separating at time m have distance valuation exactly $m\kappa$.

This gives an exact distinction between a Cantor repeller and a system with open infinitesimal fibers. The distinction is whether the expansion scale κ is an order unit of Γ , not valuation rank alone. When κ is not an order unit the iterate family is uniformly equicontinuous on the integral nonescaping set, and every compact forward-invariant subset is finite, although the scale quotient is the full shift. Every preperiodic point is a canonical center, and all preperiodicity polynomials split simply in the original Hahn field. Scalar extension can therefore create entire itinerary fibers without creating any preperiodic points. We also prove uniform support-controlled deformation estimates, an exact formula for the valuation-bounded orbit locus, and a polynomial equicontinuity theorem when the value group has no order unit. The example $F(x) = \omega(x^2 - 1)$ makes all the phenomena explicit inside **No** and **No[i]**: its centers are real surreal numbers, and the fibers in **No** are obtained by restricting to real infinitesimal errors. At the level of the whole class **No[i]**, the discreteness and compactness clauses are special cases of the collection's theorem that every set is closed and discrete in the fine topology; the substance of those clauses lies in set-sized Hahn workspaces.

Status and scope

This is an unrefereed, AI-assisted research draft with written proofs and exact finite regression tests, placed in the collection from one delivered manuscript (section 1.3). The fiber, compact-subsystem, and scalar-extension/deformation results are proposed as new results; historical priority is not certified. No named published open conjecture is claimed solved. The scale subgroup and error-ideal lemma are formalized in Lean; the dynamical theorems remain pending. See section 13.3.

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1 The problem and the contribution

A familiar construction in non-Archimedean dynamics takes finitely many inverse branches, each contracting by a fixed scale, and labels the points that never escape by infinite words. Over a rank-one valued field, the contraction scales tend to zero, and a word determines a single point. Classical quadratic Cantor Julia sets are an established example; see Benedetto's lecture notes, Example 4.39 [4].

The same conclusion need not survive in a larger Hahn field, and it does not survive unchanged in the full surreal or surcomplex field. If $\kappa > 0$ is the valuation gained by one inverse branch, then the gains $n\kappa$ may remain below a larger valuation η for every ordinary integer n . Successive choices resolve a point to every q -order, but may never resolve a perturbation of valuation η . There are two separate questions: whether every word is realized at all, and exactly how many points realize it. A topological contraction argument alone answers neither question at arbitrary rank.

We answer both questions for a substantial polynomial family, including arbitrary infinitesimal coefficient perturbations of degree at most d of a degree- d polynomial with simple split reduction. The main formula is

$$\mathcal{B}_{\text{int}}(F) = \coprod_{s \in \Sigma_d} (x_s + I_\kappa), \quad I_\kappa = \bigcap_{n \geq 0} \{h \in K : v(h) > n\kappa\}. \quad (1.1)$$

The right-hand side describes actual points, not only formal labels or an unspecified intersection of disks. The centers x_s are constructed with one support certificate valid for all itineraries. Their construction does not presume topological convergence of a sequence of Hahn elements.

The result has several consequences which are useful precisely when the rank-one picture breaks down. The preperiodic points remain rigid while the fibers grow. The native topology can make all compact invariant subsystems finite while a coarser scale quotient has entropy $\log d$. Perturbations beyond every power of q preserve the whole itinerary partition, yet move the actual periodic points. Finally, leaving the integral disk and having a valuation-unbounded orbit are different properties; we calculate exactly when they agree.

1.1 Relation to the repository at the pinned revision

The repository used is VladimirReshetnikov/Surreal, at revision

a3124af79f66b8b9c196d76b4cbc5ac3938907c4.

Its root and documentation inventories were read, along with the description of docs/surcomplex/dynamics-and-normal-forms. That report concerns local normal forms, return thresholds, small divisors, and associated formal/support questions. The present article studies a global integral nonescaping set and its infinite symbolic fibers instead [5, 6].

This was a targeted inventory audit, not a line-by-line review of every long manuscript or Lean file. The project already develops Hahn support machinery and distinctions among analytic categories. Those ideas are not presented here as discoveries. We use a classical Hahn-field workspace, state every support hypothesis, and give the polynomial and topological arguments in full. The relation to the reports of the collection as it now stands, including reports the manuscript did not consult, is recorded in [section 1.4](#).

1.2 What is, and is not, claimed to be new

The positive-support lemma, the formal implicit-function construction, valuation coarsening, and the one-sided full shift are standard tools. The rank-one Cantor construction and the elementary periodic-word counts are also standard. Our proposed new contribution is the combined arbitrary-rank theorem: exact affine itinerary fibers; the sharp order-unit/topology dichotomy; finiteness of compact forward-invariant subsets when the scale is not an order unit; and the scalar-extension and support-controlled deformation consequences. (Placement note: this claim concerns set-sized Hahn workspaces. For the whole class $\mathbf{No}[i]$ with its fine topology, the discreteness and compactness clauses are special cases of a theorem already in the collection; see [section 1.4](#) and [remark 12.3](#).)

A targeted search through public sources on Hahn fields, non-Archimedean polynomial dynamics, higher-rank valuations, and surreal analysis did not locate these statements in this form. That is not a proof of historical originality. The article therefore supplies proofs, not a claim of independently certified priority. It does not purport to resolve the unrelated birthday, automorphism, or differential-equation questions in the repository.

Three distinctions that cannot be suppressed

An element of negative derivative valuation is called *valuation-expanding* here; this does not automatically make its periodic point repelling for a real-valued absolute value. An *integral nonescaping* orbit stays in $\mathcal{O}$; a *valuation-bounded* orbit merely has some lower valuation bound. Finally, strong Hahn summation is an algebraic support operation, not necessarily a limit in the full valuation topology.

1.3 Provenance of this report

This report is built from **one** delivered manuscript, the fifth item of the eighteenth batch of manuscripts received by the collection (archive name `surreal_symbolic_dynamics`). It was delivered on 22 September 2026 as a 24-page research draft; its PDF metadata describe it as a research draft prepared with ChatGPT. The delivered title was *Expanding Polynomial Dynamics over Surreal and Surcomplex Fields: Exact symbolic fibers and the cofinality obstruction*. Its repository pin is the revision `a3124af` displayed in [section 1.1](#). The cross-references added at placement were checked against the collection at commit `e9a9650`.

This is not a merge. There was no second source, and nothing was selected out of a larger body of work: every theorem, proof, example, question, and limitation of the manuscript is kept. The placement changed the following, and nothing else.

- (1) Every label carries the prefix `epd:`.
- (2) The manuscript called an order unit a *cofinal scale* and spoke of the *cofinal* and *noncofinal* cases. That word is reserved in the collection for another invariant, so the property is called an order unit throughout. The subtitle and the name of [theorem 7.1](#) change accordingly. The dictionary is stated once, in [remark 2.2](#).
- (3) Cross-notes to neighbouring reports were added ([section 1.4](#) and [remark 12.3](#)), together with a pointer to the collection's Lean proof of the positive-support lemma ([section 13.3](#)).
- (4) The reproduction instructions were adapted to the placed file layout. The verification program overwrites its own recorded output by default, and [section 13.2](#) says how to avoid that.

(5) The non-claims scattered through the manuscript are collected in Appendix C.

At placement, no hypothesis, statement, or proof was altered beyond the change of word in (2). A subsequent proof review made degree preservation explicit in the summaries, added its counterexample below (3.1), expanded the noncompactness and formal-substitution arguments, and updated Lean coverage. The delivered code and audit remain unchanged. The report is filed under `surcomplex` although the archive name says *surreal*. Its theorems hold uniformly for $k = \mathbb{R}$ and $k = \mathbb{C}$, the explicit specialization (theorem 12.1) is stated in `No[i]`, and its valuation-theoretic neighbours are `surcomplex` reports. The real case is kept explicit throughout. For $k = \mathbb{R}$ the hypothesis (3.1) asks for real, distinct residue roots. The centers of the surreal example are real, and its fibers in `No` are obtained by restricting to real errors.

1.4 Relation to other reports in this collection

The manuscript consulted only the repository inventories and the description of the dynamics report. The following notes place it among the reports the collection now contains. They were written at placement. No result is transferred into this report from any of them, and none of their results is strengthened here.

Surcomplex Dynamics: Linearization Thresholds, Periods, and Normal Forms (`docs/surcomplex/dynamics-and-normal-forms/`). The two reports are about different problems. That report linearizes *germs* at a fixed point whose multiplier is an ordinary complex number of modulus one, with small divisors, return thresholds, and coefficient-function categories. Its limitations section excludes “global Julia or Fatou theory on the proper class `No[i]`” and, from the scope of its exact finite-return linearization theorem (`dyn:thm:exact`), “attracting or repelling multipliers” (items N51 and N18 of its non-claims list when this note was written). This report answers *neither* exclusion. It studies the global integral nonescaping set of a polynomial whose periodic points have valuation-expanding multipliers. It constructs no Julia or Fatou theory: $J_{\text{aff},v}$ of definition 11.3 is a deliberately named affine equicontinuity set, and corollary 11.4 says nothing about spherical or Berkovich Julia sets. By remark 7.2, “valuation-expanding” is not “repelling”. It linearizes nothing at any periodic point. Several letters and words mean different things in the two reports, and each keeps its own meaning:

- σ is the one-sided shift here. There $\sigma(\lambda)$ is the drifted exponential small-divisor rate.
- $\kappa = v(q) > 0$ is the valuation gained by an inverse branch here. There it is the generic divisor rate of `dyn:rem:oneinvariant`, and also appears as κ_{Υ} .
- q is a Hahn element of valuation κ here. There it is the order in the resonance tag RES_q , and q_j are continued-fraction denominators.
- A *multiplier* here is $(F^p)'(x) \in K$ at a periodic point, a Hahn element of valuation $-p\kappa < 0$ (proposition 6.2). There it is an ordinary unit complex number.
- *Julia* occurs here only in the subscripted affine set $J_{\text{aff},v}$. There it refers to Julia’s functional equation (`dyn:cor:julia`).

The quadratic critical scale of that report (`dyn:prop:quadratic-scale`) rescales $\lambda z + t^\eta z^2$ at a unit multiplier, as a boundary of a linearization domain. It is unrelated to the expanding scale q^{-1} used here.

Surcomplex Analysis (docs/surcomplex/analysis/). Its fine topology (`a:def:fine`) is the topology that the manuscript calls the full native, or order (fine), topology of $\mathbf{No}$ and $\mathbf{No}[i]$ in section 12. Its `a:thm:discrete` proves that every set $A \subseteq \mathbf{No}[i]$ is closed and discrete in that topology, with one separating radius for all pairs of points of A . The surreal-level discreteness and compactness statements of this report are therefore special cases: see remark 12.3. The substance of theorems 7.1 and 7.5 lies in set-sized workspaces $k((t^\Gamma))$ with their intrinsic valuation topologies. There a set need not be discrete, and the finiteness of compact invariant subsets is not automatic. The manuscript’s remark comparing the fine topology with a workspace topology (remark 12.2) is an instance of `a:rem:notvaluation`. The uniform equicontinuity statements over whole fine balls, which are proper classes, are not consequences of `a:thm:discrete`.

Cofinality, Factorization and Scalar Extension for Entire Hahn Functions at Arbitrary Rank (docs/surcomplex/entire-functions-at-arbitrary-rank/). An order unit in definition 2.1 is exactly an order unit in the sense of `ent:def:order-unit`. That report reserves the word *cofinality* for $\text{cf}(\Gamma)$ (`ent:rem:one-name`), which is why this report does not call an order unit a cofinal scale. The maximal proper convex subgroup H of section 10.1 is the kernel of the order-preserving homomorphism $\pi : \Gamma \rightarrow \mathbb{R}$ of `ent:lem:coarsening`. That lemma is stated under that report’s standing divisibility hypothesis. The facts about H used here are proved directly in section 10.1 and need no divisibility. The countable group without an order unit in example 11.5 is, as an ordered group, that report’s $\Gamma_\infty = \bigoplus_{j \geq 0} \mathbb{Q} \omega^j$ (`ent:subsec:gamma-infinity`). Its `ent:cor:cofinality` is a dichotomy in $\text{cf}(\Gamma)$ for entire functions. It is not the order-unit dichotomy of theorem 7.1.

Hahn–Tate Uniformization at Arbitrary Valuation Rank (docs/surcomplex/hahn-tate-uniformization/). The convex subgroup H_κ of definition 2.1 is the subgroup H_α of `tate:eq:H`, for $\alpha = \kappa$: the smallest convex subgroup containing the scale. There it bounds the domain of the Tate parametrization. Here it bounds the error ideal I_κ . No result is shared.

Rank-One Non-Archimedean Analytic Geometry inside the Surcomplex Numbers (docs/surcomplex/rank-one-berkovich/). When κ is an order unit, theorem 7.1(a) has the same symbolic Cantor model as the classical rank-one construction cited in [4, Example 4.39]. An order unit does not make Γ rank one: $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ with $\kappa = (1, 0)$ is already a counterexample. The citation provides the rank-one quadratic comparison; the proof here uses the exact cylinder valuations even at higher rank. In the fixed field $\mathbb{C}((t^\mathbb{R}))$ of that report every positive element of the Archimedean group $\mathbb{R}$ is an order unit, so only that case can occur there. That report contains no polynomial dynamics. No statement about Berkovich spaces, Julia sets, or Tate algebras is proved or used here, and no transfer in either direction is claimed.

2 Workspaces, scales, and support

2.1 The valued field

Throughout Sections 2–11, let

$$k \in \{\mathbb{R}, \mathbb{C}\}, \quad K = k((t^\Gamma)),$$

where $\Gamma \neq 0$ is a set-sized ordered abelian group. An element is a series $a = \sum_{\gamma \in S} a_\gamma t^\gamma$ with $S \subseteq \Gamma$ well ordered. All coefficients in k are trivially valued. Set

$$v(a) = \min \text{supp}(a) \quad (a \neq 0), \quad v(0) = +\infty, \quad \mathcal{O} = \{a : v(a) \geq 0\}, \quad \mathfrak{m} = \{a : v(a) > 0\}.$$

The residue map $\text{st} : \mathcal{O} \rightarrow k$ extracts the coefficient at 0. The usual complex modulus of a coefficient is not the valuation used in this article. No divisibility assumption on Γ , and no algebraic closedness of K , is needed for the main theorem.

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. The native valuation topology has balls $\{y : v(y - x) > \gamma\}$, for $\gamma \in \Gamma$, as a neighborhood basis. The associated additive uniformity has entourages $\{(x, y) : v(x - y) > \gamma\}$. These definitions make sense without a real-valued metric.

2.2 An individual scale, not the rank

Definition 2.1. A positive $u \in \Gamma$ is an *order unit* if for every $\gamma \in \Gamma$ there is an $n \in \mathbb{N}$ with $\gamma \leq nu$. For $\kappa > 0$, define

$$H_\kappa = \{\gamma \in \Gamma : |\gamma| \leq n\kappa \text{ for some } n \in \mathbb{N}\}, \quad I_\kappa = \{h \in K : v(h) > n\kappa \text{ for every } n \in \mathbb{N}\}.$$

The convention $v(0) = +\infty$ includes zero in I_κ . When the scale $\kappa = v(q)$ of a map is an order unit we speak of the *order-unit case*. Otherwise we speak of the *non-order-unit case*.

Remark 2.2 (One name for one property). The delivered manuscript called an order unit a *cofinal scale*. It called the two cases the *cofinal* and *noncofinal* cases, and the distinction the *cofinality* of the expansion scale. In this collection, the report `entire-functions-at-arbitrary-rank` (`ent:rem:one-name`) reserves *cofinality* for $\text{cf}(\Gamma)$, the least cardinality of a cofinal subset of Γ . Its `ent:def:order-unit` is the notion defined above. This report therefore uses the following dictionary, and only it.

Delivered manuscript	This report
κ is cofinal; a cofinal scale	κ is an order unit; an order-unit scale
κ is noncofinal	κ is not an order unit
the cofinal / noncofinal case	the order-unit / non-order-unit case
cofinality dichotomy (theorem 7.1)	order-unit dichotomy
cofinality is lost (theorem 8.1)	the order-unit property is lost

The word *cofinal* survives only in its subset sense. A subset S of Γ (or of $\Gamma_{>0}$) is cofinal if every element is at most some member of S . Thus $u > 0$ is an order unit exactly when $\{nu : n \in \mathbb{N}\}$ is cofinal in Γ , in agreement with `docs/NOTATION.md`. Apart from the quotations of the manuscript in this remark and in [section 1.3](#), wherever *cofinality* occurs in this report it means $\text{cf}(\Gamma)$. It occurs in [example 11.5](#) and in names taken from the `entire-functions` report. The delivered file `RESEARCH_STATUS.md`, kept unchanged, still uses the manuscript's words.

Lemma 2.3. H_κ is a convex subgroup of Γ . The set I_κ is an ideal of $\mathcal{O}$, and, for any q with $v(q) = \kappa$, one has $qI_\kappa = I_\kappa$. Moreover

$$I_\kappa = \bigcap_{n \geq 0} q^n \mathcal{O}, \quad I_\kappa = 0 \iff \kappa \text{ is an order unit}.$$

If κ is not an order unit, I_κ is a nonzero clopen additive subgroup of K . It is the maximal ideal of the valuation ring for the coarsened valuation $K^\times \rightarrow \Gamma/H_\kappa$.

Proof. Closure and convexity of H_κ follow by adding the finite bounds. The valuation inequality for a sum and the inequality $v(ah) \geq v(h)$ for $a \in \mathcal{O}$ prove the ideal assertion. Adding or subtracting κ preserves the condition of exceeding every $n\kappa$, proving $qI_\kappa = I_\kappa$. Requiring $v(h) \geq n\kappa$ for all n is equivalent to requiring $v(h) > n\kappa$ for all n , by increasing n by one. This proves the intersection formula.

If κ is an order unit, no finite valuation can exceed all $n\kappa$. Otherwise there exists $\eta \in \Gamma$ with $\eta > n\kappa$ for all n , and $t^\eta \in I_\kappa \setminus \{0\}$. The ball $\{v > \eta\}$ is contained in I_κ , so this subgroup is open, and every open subgroup of a topological group is closed. Finally, $v(h) + H_\kappa > 0$ means precisely that $v(h)$ exceeds every element of H_κ , equivalently every $n\kappa$. $\square$

The ideal is thus not an ad hoc error class. It is the exact error left unresolved by the convex subgroup generated by the dynamical scale. It is generally an ideal of a valuation ring, not an ideal of the field K .

Example 2.4. In $\Gamma = \mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$, the scale $(0, 1)$ is not an order unit, since $(1, 0) > n(0, 1)$ for every n . In the very same group, $(1, 0)$ is an order unit. Consequently “rank greater than one” cannot replace “the chosen scale is not an order unit” in any theorem below.

2.3 The summation tool and its exact role

A family $(a_j)_{j \in J}$ of Hahn series is *strongly summable* if its support union is well ordered and every exponent occurs in only finitely many members of the family. Its sum is then defined coefficientwise.

We use the classical positive-support lemma associated with Neumann’s ordered-series construction [1]: if $S \subseteq \Gamma_{>0}$ is well ordered, the additive monoid $\langle S \rangle$ is well ordered, and each fixed element of Γ has only finitely many representations as a finite word in S , allowing all finite word lengths. Repetitions and ordering of letters are allowed in this finiteness assertion. The following useful form spells out the only infinite evaluation operation needed here.

Lemma 2.5 (Finite-parameter Hahn evaluation). *Let $a_1, \dots, a_r \in \mathfrak{m}$. Every formal series $A \in k[[Z_1, \dots, Z_r]]$ has a well-defined strong evaluation $A(a_1, \dots, a_r) \in \mathcal{O}$. Evaluation is a k -algebra homomorphism. Every evaluated support is contained in*

$$M = \left\langle \bigcup_{j=1}^r \text{supp}(a_j) \right\rangle.$$

Evaluation respects formal substitution by series with zero constant term. If a formal series belongs to an ideal $(Z_{j_1}, \dots, Z_{j_s})$, its evaluation belongs to the ideal $(a_{j_1}, \dots, a_{j_s})\mathcal{O}$.

Proof. The finite support union is well ordered and positive. Expand every parameter monomial into words in this union, marking a letter by the parameter from which it came. The positive-support lemma bounds both the possible words and their lengths at each output exponent. There are only finitely many choices of a parameter mark at every letter. Thus the expanded family has well-ordered support and finite coefficient fibers. Addition, multiplication, and zero-constant formal substitution may all be regrouped coefficientwise, because at each exponent only finitely many terms participate.

For the ideal assertion, write the formal series as a finite sum $\sum_{\ell=1}^s Z_{j_\ell} B_\ell$. Each B_ℓ evaluates into $\mathcal{O}$ by the first assertion. Apply multiplicativity. $\square$

Remark 2.6. The lemma does not assert that partial sums converge in the native topology. For instance, in the rank-two example with $q = t^{(0,1)}$, $\sum_{n \geq 0} q^n = (1 - q)^{-1}$ is a valid Hahn identity,

while the remainders have valuations $(0, N)$ and never enter the neighborhood $\{v > (1, 0)\}$. Algebraic identities must be established before parameter evaluation; an unproved continuity assertion cannot replace that step.

3 Simple reduction and universal inverse branches

Fix $q \in \mathfrak{m} \setminus \{0\}$, write $\kappa = v(q)$, and let

$$P(X) = P_0(X) + \sum_{j=0}^d e_j X^j \in \mathcal{O}[X], \quad e_j \in \mathfrak{m}, \quad P_0(X) = a_d \prod_{i=1}^d (X - c_i), \quad (3.1)$$

where $d \geq 2$, $a_d \in k^\times$, and the $c_i \in k$ are distinct. Zero perturbation coefficients are permitted. In particular P has degree d and unit leading coefficient. The equality $\deg P = \deg P_0$ is essential for the exterior-orbit and global preperiodicity claims. For example, over $\mathbb{R}((t^{\mathbb{Z}}))$ take

$$q = t, \quad P(X) = (X^2 - 1)(1 + tX).$$

Its reduction still has the two simple roots ± 1 , but $\deg P = 3$. For $F = t^{-1}P$, one has $F(-t^{-1}) = 0$ and $F(0) = -t^{-1}$. This gives a two-cycle with a nonintegral point, so the claims below cannot be extended merely by allowing infinitesimal terms of higher degree.

Primes below always denote formal differentiation with respect to X , holding coefficients fixed; no differential-field derivation on the surreal coefficients is used. Set

$$F(X) = q^{-1}P(X), \quad \mathcal{B}_{\text{int}}(F) = \{x \in K : F^n(x) \in \mathcal{O} \text{ for all } n \in \mathbb{N}\}.$$

For $x \in \mathcal{B}_{\text{int}}(F)$, the identity $P(F^n(x)) = qF^{n+1}(x)$ implies $P_0(\text{st}(F^n(x))) = 0$. Thus there is a well-defined itinerary

$$\pi(x) = (s_0, s_1, \dots) \in \Sigma_d := \{1, \dots, d\}^{\mathbb{N}}, \quad \text{st}(F^n(x)) = c_{s_n}.$$

We write $\sigma(s_0, s_1, \dots) = (s_1, s_2, \dots)$.

Lemma 3.1 (Universal implicit branch). *For each i , there exists a unique $\Psi_i \in k[[E_0, \dots, E_d, V]]$ such that*

$$\Psi_i(0, 0) = c_i, \quad P_0(\Psi_i) + \sum_{j=0}^d E_j \Psi_i^j = V. \quad (3.2)$$

Moreover, there is a universal $A_i \in k[[E_0, \dots, E_d, U, W]]$ for which

$$\Psi_i(E, U) - \Psi_i(E, W) = (U - W)A_i(E, U, W), \quad (3.3)$$

where A_i is a unit with constant term $1/P'_0(c_i)$. The identity may be substituted into another formal power-series ring whenever all substituted parameters have positive total order.

Proof. Put $\Psi_i = c_i + Y$, with Y of positive total degree. At total degree $r \geq 1$, the only occurrence of the unknown degree- r part of Y is its product with $P'_0(c_i)$. Every other term uses already determined lower-degree parts, or has a positive-degree parameter as an additional factor. Since $P'_0(c_i) \neq 0$, this determines Y recursively and proves existence and uniqueness.

Subtract the two equations in (3.2). The polynomial divided difference in the two branch values has constant term $P'_0(c_i)$, hence is a unit. Its inverse gives (3.3). $\square$

Proposition 3.2 (Exact branch metric). *For every $w \in \mathcal{O}$ and every i , there is a unique $y \in \mathcal{O}$ with $\text{st}(y) = c_i$ and $P(y) = qw$. Denote it by $\Phi_i(w)$. For all $w, z \in \mathcal{O}$,*

$$v(\Phi_i(w) - \Phi_i(z)) = \kappa + v(w - z). \quad (3.4)$$

If $x, y \in \mathcal{O}$ have the same residue c_i , then

$$v(F(x) - F(y)) = v(x - y) - \kappa. \quad (3.5)$$

Equal inputs are interpreted using $+\infty$.

Proof. Evaluate $\Psi_i(E, V)$ at $(e_0, \dots, e_d, qw)$, all nonzero inputs having positive valuation. **Lemma 2.5** proves existence and the defining equation. The residue is c_i .

For uniqueness and the exact metric, use the polynomial identity

$$P(Y) - P(Z) = (Y - Z)D_P(Y, Z).$$

When both arguments have residue c_i , the residue of D_P is $P'_0(c_i) \neq 0$, so $v(D_P) = 0$. Substituting $Y = \Phi_i(w)$ and $Z = \Phi_i(z)$ gives $v(Y - Z) = v(q(w - z))$. Substituting arbitrary x, y in the same residue class gives (3.5). It also proves that two lifts of the same w coincide. $\square$

In particular, F maps each of its d integral inverse-branch pieces bijectively onto $\mathcal{O}$. Different pieces have points whose differences have valuation 0. The branch calculation is algebraic; spherical completeness or a Banach fixed-point theorem has not been used.

4 Canonical realization of every itinerary

Introduce the finite-parameter ring

$$R = k[[Q, E_0, \dots, E_d]], \quad \mathcal{P}(X) = P_0(X) + \sum_{j=0}^d E_j X^j.$$

For $U \in R$, let $\Phi_i(U) = \Psi_i(E, QU)$. Substitution is valid since QU has zero constant term. By **lemma 3.1**,

$$\Phi_i(U) - \Phi_i(W) \in Q(U - W)R. \quad (4.1)$$

For $s \in \Sigma_d$, form the depth- N expression

$$X_s^{[N]} = \Phi_{s_0} \circ \dots \circ \Phi_{s_{N-1}}(0), \quad X_s^{[0]} = 0.$$

Theorem 4.1 (Universal centers and common support). *There is a formal series $X_s \in R$ for every $s \in \Sigma_d$ with*

$$X_s - X_s^{[N]} \in Q^N R, \quad \mathcal{P}(X_s) = QX_{\sigma s}, \quad X_s(0) = c_{s_0}. \quad (4.2)$$

After evaluation $Q = q, E_j = e_j$, set $x_s = \text{ev}(X_s)$. Then

$$F(x_s) = x_{\sigma s}, \quad \pi(x_s) = s, \quad (4.3)$$

and, simultaneously for all s ,

$$\text{supp}(x_s) \subseteq M := \left\langle \text{supp}(q) \cup \bigcup_{j=0}^d \text{supp}(e_j) \right\rangle. \quad (4.4)$$

Every coefficient of total parameter degree r in X_s depends only on $s_0, \dots, s_r$.

Proof. Repeated use of (4.1) shows that for $L \geq N$,

$$X_s^{[L]} - X_s^{[N]} \in Q^N R.$$

The ring R is Q -adically complete and separated: grouping a formal series by its Q -degree identifies it with $k[[E_0, \dots, E_d]][[Q]]$. For each fixed Q -degree, the entire coefficient series in E stabilizes once N exceeds that degree. These stabilized coefficients define a unique X_s , with the displayed congruence for every N . This completion takes place in an ordinary finite-variable formal ring, not in the valuation topology of K .

The finite expressions satisfy $\mathcal{P}(X_s^{[N+1]}) = QX_{\sigma s}^{[N]}$. Passing to their formal Q -adic limits proves the middle identity in (4.2). The other two assertions follow directly from the construction. If $N = r + 1$, the difference is divisible by Q^{r+1} , so every coefficient of total degree at most r has already stabilized and only the first $r + 1$ symbols have been used.

Evaluate the established formal polynomial identity by lemma 2.5. This proves (4.3). Reduction at each shifted word gives the itinerary. The same evaluation lemma gives (4.4), with no dependence of M on s . $\square$

The formal section is distinguished by the fixed coefficient embedding $k \hookrightarrow K$, the polynomial reduction P_0 , and the displayed universal construction. We do not assert an intrinsic uniqueness among all possible set-theoretic sections of π .

Corollary 4.2 (Constant coefficients). *If $P = P_0$ and $q = t^\kappa$, then each x_s has the form $\sum_{n \geq 0} a_n(s)t^{n\kappa}$ with $a_n(s) \in k$. The coefficient $a_n(s)$ depends only on the first $n + 1$ symbols. The map $s \mapsto x_s$ is unchanged when the exponent group is enlarged.*

Proof. With all $E_j = 0$, X_s belongs to $k[[Q]]$. Evaluate coefficientwise. The same formal series is evaluated under every compatible Hahn-field embedding. $\square$

Remark 4.3 (A limit that is not taken). For $P = X^2 - 1$, label roots by signs ± 1 . The depth- N nested branch with terminal zero differs from x_s by an element of valuation exactly $N\kappa$: apply (3.4) N times to the tail $x_{\sigma^N s}$ and 0, whose residues differ. When κ is not an order unit, these approximants do not converge to x_s in the native valuation topology. Nevertheless x_s is a perfectly well-defined Hahn element by the preceding theorem.

5 The exact symbolic-fiber theorem

Theorem 5.1 (Exact fibers and exact separation). *Under (3.1), the itinerary map is onto and*

$$\pi^{-1}(s) = x_s + I_\kappa, \quad \mathcal{B}_{\text{int}}(F) = \prod_{s \in \Sigma_d} (x_s + I_\kappa). \quad (5.1)$$

For $h \in I_\kappa \setminus \{0\}$ and every $n \geq 0$,

$$v(F^n(x_s + h) - x_{\sigma^n s}) = v(h) - n\kappa. \quad (5.2)$$

If $s \neq r$ first differ at index m , then for every $x \in \pi^{-1}(s)$ and $y \in \pi^{-1}(r)$,

$$v(x - y) = m\kappa. \quad (5.3)$$

Thus a word determines a unique point if and only if κ is an order unit in the ambient value group.

Proof. Surjectivity is already supplied by the centers. Let $h \in I_\kappa$. Initially $x_s + h$ and x_s have the same residue. If their n th iterates have difference valuation $v(h) - n\kappa$, that valuation is positive by the definition of I_κ . They therefore have the same residue at time n , and (3.5) proves the next instance of (5.2). Induction proves both that the perturbed point remains integral and that its itinerary is s .

Conversely, let x have itinerary s . For every N we may recover x and x_s by applying the same first N inverse branches to $F^N(x)$ and $x_{\sigma^N s}$, respectively. These two terminal points are integral, so their difference has nonnegative valuation. Applying (3.4) N times gives

$$v(x - x_s) \geq N\kappa.$$

This holds for every N , and lemma 2.3 implies $x - x_s \in I_\kappa$. No limiting argument is involved.

For two different itineraries, their m th iterates have distinct residues and hence difference valuation 0. Pulling back along their common m branches gives (5.3). In particular the cosets in (5.1) are disjoint. The last assertion follows from lemma 2.3. $\square$

Corollary 5.2 (Exact finite cylinders). *For $x, y \in \mathcal{B}_{\text{int}}(F)$ and $N \geq 0$, the first N itinerary symbols agree if and only if $v(x - y) \geq N\kappa$.*

Proof. A first mismatch $m < N$ gives valuation $m\kappa < N\kappa$. If the first N symbols agree, either the first mismatch is later, giving the required inequality, or the itineraries are equal and their difference belongs to I_κ . $\square$

Corollary 5.3 (The action on the fibers). *F maps $x_s + I_\kappa$ bijectively onto $x_{\sigma s} + I_\kappa$. Every point of $\mathcal{B}_{\text{int}}(F)$ has exactly d preimages in $\mathcal{B}_{\text{int}}(F)$, one in each allowable preceding-symbol fiber.*

Proof. Forward inclusion is (5.2). Given a point in the target fiber, apply Φ_{s_0} . Its itinerary is s , by uniqueness of the inverse branch, and it is the only preimage in that residue class. There are d residue classes and F has degree d . $\square$

This theorem identifies the information content of a full infinite itinerary. It determines every finite q -scale but, in general, not a point. The unresolved ambiguity is exactly an affine ideal, not an unspecified smaller disk and not a choice of a formal completion.

6 Periodic and preperiodic rigidity

Theorem 6.1 (All preperiodicity equations split). *A point of $\mathcal{B}_{\text{int}}(F)$ is preperiodic if and only if it is x_s for an eventually periodic word s . For every $m \geq 0$ and $p \geq 1$, the polynomial*

$$F^{m+p}(X) - F^m(X) \tag{6.1}$$

has exactly d^{m+p} distinct roots, all in K and all in $\mathcal{B}_{\text{int}}(F)$. They are precisely the centers with $\sigma^{m+p}s = \sigma^m s$. In particular

$$\# \text{Fix}(F^p) = d^p.$$

There are no preperiodic points outside $\mathcal{O}$.

Proof. First consider a periodic word s of period dividing p . Its center is fixed by F^p . If $x = x_s + h$ with $h \neq 0$ in I_κ were also fixed, then (5.2) would imply simultaneously

$$v(F^p(x) - F^p(x_s)) = v(h) - p\kappa \quad \text{and} \quad v(F^p(x) - F^p(x_s)) = v(h),$$

a contradiction. Hence the center is the unique periodic point in its fiber. An eventually periodic point maps to a periodic point after some m iterations. Its difference from the corresponding center never becomes zero, by (5.2), unless it was zero initially. This proves the first assertion; the converse follows from (4.3).

There are d^{m+p} words satisfying $\sigma^{m+p}s = \sigma^m s$: prescribe the first $m + p$ symbols, and repeat the last p thereafter. They give distinct roots of (6.1). That polynomial has degree d^{m+p} , since $d \geq 2$ and $p \geq 1$. Thus they are all its roots, and it splits with simple roots in K .

For completeness, if $v(x) < 0$, the leading term of $P(x)$ strictly dominates every lower-degree term. Consequently

$$v(F(x)) = d v(x) - \kappa < v(x) < 0. \quad (6.2)$$

Such a point never returns to $\mathcal{O}$ and cannot be preperiodic, because its successive valuations strictly decrease. $\square$

Proposition 6.2 (Exact multipliers). *For $x \in \mathcal{B}_{\text{int}}(F)$ and $n \geq 1$,*

$$v((F^n)'(x)) = -n\kappa.$$

At a root of (6.1), its derivative has valuation $-(m + p)\kappa$.

Proof. At every integral orbit point, P' has residue $P'_0(c_{s_j}) \neq 0$. Thus $v(F') = -\kappa$, and the chain rule gives the first formula. At a root x of (6.1), factor its derivative as

$$(F^m)'(x) ((F^p)'(F^m(x)) - 1).$$

The second factor has valuation $-p\kappa$, since that is negative and $v(1) = 0$. This proves the claimed valuation and directly confirms simplicity. $\square$

Corollary 6.3 (Periodic counts and zeta series). *The number of points of exact period p is*

$$N_p = \sum_{e|p} \mu(p/e) d^e,$$

and the number of cycles of exact period p is N_p/p . The formal Artin–Mazur zeta series is

$$\exp\left(\sum_{p \geq 1} \frac{\# \text{Fix}(F^p)}{p} z^p\right) = \frac{1}{1 - dz}.$$

Proof. Every point fixed by F^p has a unique exact period dividing p . Möbius inversion of $d^p = \sum_{e|p} N_e$ gives the first formula. Each cycle has p points. The last identity is the formal identity $\sum_{p \geq 1} (dz)^p / p = -\log(1 - dz)$. $\square$

The formulas themselves are classical full-shift formulas. Their role here is to show what periodic data *fails* to detect: the same counts and multipliers persist whether the nonescaping set consists of individual centers or of open infinitesimal fibers.

7 The native topology and the scale quotient

Give Σ_d the product topology of the finite discrete alphabet. Its cylinder of depth N fixes the first N symbols.

Theorem 7.1 (Order-unit dichotomy). *The following conclusions hold.*

- (a) If κ is an order unit, then $I_\kappa = 0$ and $s \mapsto x_s$ is a homeomorphism $\Sigma_d \rightarrow \mathcal{B}_{\text{int}}(F)$, conjugating the one-sided shift to F . The set $\mathcal{B}_{\text{int}}(F)$ is compact and perfect. Periodic points are dense in it. The family $(F^n)_{n \geq 0}$ is not equicontinuous at any point of $\mathcal{B}_{\text{int}}(F)$, even relative to that set.
- (b) If κ is not an order unit, then $\mathcal{B}_{\text{int}}(F)$ is clopen and noncompact in K , every fiber $x_s + I_\kappa$ is open, and every subset of $C := \{x_s : s \in \Sigma_d\}$ is closed and discrete in K . The itinerary map $\pi : \mathcal{B}_{\text{int}}(F) \rightarrow \Sigma_d$ is continuous but is not a quotient map. The native topological quotient by the fibers is discrete.
- (c) In the non-order-unit case the family $(F^n)_{n \geq 0}$ is uniformly equicontinuous on $\mathcal{B}_{\text{int}}(F)$ in the native valuation uniformity.

Proof. When κ is an order unit, [corollary 5.2](#) and the order-unit property identify the cylinder basis with a neighborhood basis on $\mathcal{B}_{\text{int}}(F)$. The center map and its inverse are continuous. Compactness and perfectness follow from the elementary corresponding properties of the finite-alphabet product. A periodic word extending any finite prefix proves density. For failure of equicontinuity, fix any x_s and any input neighborhood. Choose a word r sharing a sufficiently long prefix with s but first differing at a later index m . The input difference has valuation $m\kappa$, arbitrarily large because κ is an order unit; the difference after m iterations has valuation 0. The fixed output neighborhood $\{v > 0\}$ cannot be maintained.

Now choose $\eta > n\kappa$ for every n . The fibers are open by [lemma 2.3](#), so their union is open. Also $\mathcal{B}_{\text{int}}(F) = \bigcap_{n \geq 0} (F^n)^{-1}(\mathcal{O})$ is closed, since $\mathcal{O}$ is closed and polynomials are continuous. Distinct centers have difference valuation $m\kappa < \eta$. Thus a ball of threshold η contains at most one point of any subset $A \subseteq C$, proving discreteness. To prove closedness, a point in the closure of A has such a ball meeting A in exactly one point a . Unless the original point equals a , a smaller ball excludes a , contradicting closure. Hence A is closed. In particular C is an infinite closed discrete subset of $\mathcal{B}_{\text{int}}(F)$. If $\mathcal{B}_{\text{int}}(F)$ were compact, so would be C , contradicting the open cover of C by its singletons. This proves noncompactness.

By [corollary 5.2](#), every cylinder preimage is open, so π is continuous. Each individual fiber is open, whereas a singleton in Σ_d is not open. Consequently π is not a quotient map. The quotient topology by open fibers is discrete.

Finally, fix an arbitrary desired output threshold $\gamma \in \Gamma$ and set

$$\delta = \max(\gamma, 0) + \eta.$$

If $x, y \in \mathcal{B}_{\text{int}}(F)$ have $v(x - y) > \delta$, their difference lies in I_κ , so they have the same itinerary by [theorem 5.1](#). Repeated use of the exact forward metric yields

$$v(F^n(x) - F^n(y)) = v(x - y) - n\kappa > \max(\gamma, 0) + \eta - n\kappa > \gamma$$

for every n . The same input threshold works for every pair and every iterate, which is uniform equicontinuity. $\square$

Remark 7.2 (Why this is not a contradiction). The valuation of a nonzero difference within a fiber strictly decreases by κ at every step. Uniform equicontinuity nevertheless holds because an input error can be placed beyond *all* the scales that this countable iteration can spend. A negative multiplier valuation therefore does not by itself establish topological repulsion. This is why the article does not identify its expanding points with classical Julia points without naming a topology and a valuation scale.

7.1 The correct full-shift quotient

Definition 7.3. On $\mathcal{B}_{\text{int}}(F)$, the κ -scale uniformity is generated by

$$E_N = \{(x, y) : v(x - y) \geq N\kappa\}, \quad N \in \mathbb{N}.$$

Its indistinguishability relation is $x - y \in I_\kappa$.

Each E_N is an equivalence relation on $\mathcal{B}_{\text{int}}(F)$, by the ultrametric inequality. The family defines a uniformity even when it is not Hausdorff.

Proposition 7.4. *The Hausdorff quotient of the κ -scale uniformity is uniformly isomorphic to Σ_d with its cylinder uniformity, and the induced map is the full shift. In particular its topological entropy is $\log d$. In the non-order-unit case this is not the native topological quotient.*

Proof. By corollary 5.2, E_N is exactly equality of the first N symbols. By theorem 5.1, the common equivalence relation of all E_N is equality of the entire word, or equivalently difference in I_κ . The quotient therefore has precisely the cylinder uniformity. The shift identity follows from the definition of itinerary.

For entropy, there are exactly d^N orbit names of length N at the resolution which distinguishes individual symbols. Refining to a fixed cylinder length r changes this count to d^{N+r-1} , without changing the exponential rate. Hence the rate is $\log d$. $\square$

7.2 Compact invariant sets disappear at the finer scale

Theorem 7.5 (Compact-subsystem rigidity). *Suppose κ is not an order unit. Every compact subset $A \subseteq \mathcal{B}_{\text{int}}(F)$ satisfying $F(A) \subseteq A$ is finite. Every point of A is a preperiodic center. Thus every compact forward-invariant subsystem has topological entropy zero, although the scale quotient in proposition 7.4 has entropy $\log d$.*

Proof. The open fiber partition covers A . Compactness implies that only finitely many fibers meet A . Since $F(A) \subseteq A$, the shift orbit of each itinerary occurring in A lies in a finite set. Each such itinerary is therefore eventually periodic.

Suppose $x = x_s + h \in A$ with $h \neq 0$. Choose $m \geq 0$ and $p \geq 1$ so that $\sigma^{m+p}s = \sigma^m s$, and put $c = x_{\sigma^m s}$. For $j \geq 0$ let $y_j = F^{m+jp}(x) \in A$. The corresponding center is always c , and theorem 5.1 gives

$$v(y_j - c) = v(h) - (m + jp)\kappa.$$

These valuations strictly decrease. For $j > i$, the two valuations are unequal, so

$$v(y_j - y_i) = v(h) - (m + jp)\kappa \leq v(h) - m\kappa.$$

The infinite set $\{y_j : j \geq 0\}$ is uniformly separated. The same ball argument used in theorem 7.1 makes it closed and discrete in K . It would be a closed subset of compact A , hence compact, but an infinite discrete space is not compact. This contradiction proves $h = 0$ for every $x \in A$.

Thus $A \subseteq C$. Every subset of C is discrete, so compactness again forces A to be finite. Forward invariance of a finite set makes all its points preperiodic. A self-map of a finite space has entropy zero. $\square$

One must allow finite preperiodic tails in this theorem: the hypothesis is $F(A) \subseteq A$, not $F(A) = A$. No conclusion that every point is periodic is intended. There is also no conflict with entropy monotonicity for compact factors: the native source space here is noncompact, and the full shift is the quotient of a specifically coarsened uniformity.

8 Scalar extension: new fibers, no new preperiodic points

Let $\Gamma \hookrightarrow \Delta$ be an order-preserving group embedding, and use the induced coefficient-preserving Hahn embedding

$$K = k((t^\Gamma)) \hookrightarrow L = k((t^\Delta)).$$

The parameters P, q are unchanged. The distinction between the two ambient fields is real even if the polynomial coefficients occupy a very small common subfield.

Theorem 8.1 (Scalar-extension theorem). *The universal centers over K and over L are the same elements of $K \subseteq L$. Moreover,*

$$\mathcal{B}_{\text{int}}(F; L) = \bigcup_{s \in \Sigma_d} (x_s + I_\kappa(L)), \quad (8.1)$$

$$\mathcal{B}_{\text{int}}(F; L) \cap K = \mathcal{B}_{\text{int}}(F; K), \quad I_\kappa(L) \cap K = I_\kappa(K). \quad (8.2)$$

Every periodic and preperiodic point over any field extension of K already belongs to K . If κ is an order unit of Γ but not of Δ , the following additional conclusions hold:

- (i) $\mathcal{B}_{\text{int}}(F; K)$ is the compact center set in the intrinsic topology of K , but it is closed and discrete in the intrinsic topology of L .
- (ii) Every point of $\mathcal{B}_{\text{int}}(F; L) \setminus \mathcal{B}_{\text{int}}(F; K)$ is transcendental over K .
- (iii) The restriction of the topology of L to K is not the intrinsic valuation topology of K .

Proof. The two evaluations of each universal formal series agree coefficientwise. All supports and all finite coefficient sums are preserved by the ordered exponent embedding. Apply theorem 5.1 in L to get (8.1). The value comparison with $n\kappa$ is unchanged on K , proving the ideal intersection. Integral membership and every polynomial iterate are also unchanged on K , which proves the nonescaping intersection.

For any field extension M/K , a preperiodic element is a root of $F^{m+p} - F^m$ for some m, p . By theorem 6.1, that polynomial already splits into distinct linear factors in $K[X]$. A root in M therefore equals one of those roots in K . This part does not require M to be a Hahn field or even to carry a valuation.

Assume now that the order-unit property is lost. The first conclusion is theorem 7.1 in the two fields. For the second, write a new point as $x_s + h$ with $h \in I_\kappa(L) \setminus \{0\}$. Because κ is an order unit of Γ , $v(h)$ exceeds every element of Γ . Suppose a nonzero polynomial $\sum_{j=0}^r b_j h^j = 0$, with $b_j \in K$, existed. Let j_0 be its least index with $b_{j_0} \neq 0$. For $j > j_0$ with $b_j \neq 0$,

$$v(b_j h^j) - v(b_{j_0} h^{j_0}) = v(b_j) - v(b_{j_0}) + (j - j_0)v(h) > 0.$$

Thus the j_0 term is the unique term of least valuation and cannot cancel. This contradiction proves h , hence $x_s + h$, transcendental over K .

Finally, on the center set the topology from K is a non-discrete Cantor topology, whereas that from L is discrete. They cannot agree on all of K . $\square$

Example 8.2 (A minimal extension that changes the dynamics). Take

$$K = \mathbb{C}((q)), \quad L = \mathbb{C}((t^{\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}})), \quad q \mapsto t^{(0,1)}, \quad F = q^{-1}(X^2 - 1).$$

Here K is identified with the Hahn subfield with exponent subgroup $\{0\} \oplus \mathbb{Z}$. The integral nonescaping set in K is the binary center set. In L , each center acquires the fiber $x_s + I_{(0,1)}$; for example $x_s + t^{(1,0)}$ is a new nonescaping point. None of these new points is algebraic over K . Nevertheless every solution of every preperiodicity equation was already in K .

The theorem separates two operations that should not be conflated: extending the field and taking a symbolic or topological model of the nonescaping set. The algebraic equations for all finite preperiodic orbits are stable under extension, while the infinite itinerary realization problem is not pointwise stable.

9 Support-controlled deformations and invisible changes

Keep q fixed. Let $\tilde{P} = P + D$, where

$$D(X) = \sum_{j=0}^d d_j X^j, \quad d_j \in \mathfrak{m}, \quad \tilde{F} = q^{-1} \tilde{P}.$$

Both polynomials have the same labeled simple reduction P_0 . Write x_s and $\tilde{x}_s$ for their respective universal centers.

Theorem 9.1 (Uniform deformation estimate). *If $D \neq 0$ and $\theta = \min_{d_j \neq 0} v(d_j) > 0$, then for every itinerary s ,*

$$v(\tilde{x}_s - x_s) \geq \theta. \quad (9.1)$$

The supports of all differences belong to the monoid generated by

$$\text{supp}(q) \cup \bigcup_j \text{supp}(e_j) \cup \bigcup_j \text{supp}(d_j).$$

Let $D_\theta(X) = \sum_j ([t^\theta] d_j) X^j \in k[X]$. The first possible correction is exactly

$$[t^\theta](\tilde{x}_s - x_s) = -\frac{D_\theta(c_{s_0})}{P'_0(c_{s_0})}. \quad (9.2)$$

If the numerator is nonzero, equality holds in (9.1). If it vanishes, cancellation is allowed; no equality is asserted.

Proof. Introduce further formal variables $D_0, \dots, D_d$. In the finite-variable power-series ring in Q, E, D , the difference

$$X_s(Q, E + D) - X_s(Q, E)$$

belongs to $(D_0, \dots, D_d)$. Indeed its specialization at $D = 0$ is zero, so each monomial has at least one D factor. Apply lemma 2.5 at $Q = q, E = e, D = d$. The two evaluations are exactly $\tilde{x}_s$ and x_s , by compatibility with formal substitution. Each difference is a finite sum $\sum_j d_j B_{s,j}$ with $B_{s,j} \in \mathcal{O}$. This proves the uniform valuation bound. The support statement is the same evaluation lemma.

Put $\Delta_s = \tilde{x}_s - x_s$. Subtract the two shift equations:

$$P(x_s + \Delta_s) - P(x_s) + D(x_s + \Delta_s) = q \Delta_{\sigma s}. \quad (9.3)$$

The right side has valuation at least $\kappa + \theta > \theta$. Every term of degree at least two in Δ_s has valuation greater than θ . At exponent θ , the first difference on the left is therefore $P'_0(c_{s_0})[t^\theta] \Delta_s$. At the same exponent, the second term is $D_\theta(c_{s_0})$. Extracting this coefficient in (9.3) proves (9.2). $\square$

Remark 9.2 (Why a formal ideal argument is necessary). The threshold θ may exceed all $n\kappa$. Then estimates for finite nested approximants with error valuation $n\kappa$ do not justify a claim about the θ coefficient of an alleged limit. The proof instead establishes divisibility by the perturbation ideal *before* evaluating. This is the same support-versus-topology issue that required the universal construction of the centers.

Corollary 9.3 (Deformations invisible at all dynamical orders). *If every coefficient d_j belongs to I_κ , then*

$$\mathcal{B}_{\text{int}}(F) = \mathcal{B}_{\text{int}}(\tilde{F}), \quad \pi_F^{-1}(s) = \pi_{\tilde{F}}^{-1}(s) \quad (s \in \Sigma_d).$$

The two maps induce the same full shift on the scale quotient. For any nonzero constant perturbation $D = \varepsilon \in I_\kappa$, however, F and $\tilde{F}$ have no common fixed point.

Proof. There are finitely many coefficients. If $D \neq 0$, their minimum nonzero valuation still exceeds every $n\kappa$. [Theorem 9.1](#) therefore gives $\tilde{x}_s - x_s \in I_\kappa$ for every s . The fiber formula makes the labeled cosets identical. Both quotient actions shift the labels. Alternatively, $D(x)/q \in I_\kappa$ for integral x because $qI_\kappa = I_\kappa$. If a common fixed point existed for a nonzero constant D , it would satisfy both $P(x) = qx$ and $P(x) + \varepsilon = qx$, which is impossible. $\square$

Thus even the entire infinite symbolic partition, together with all finite-scale data, does not determine the actual fixed points in a workspace in which κ is not an order unit. The correction formula shows exactly where the missing information first appears.

10 Integral nonescape versus valuation-bounded orbits

Define the other natural orbit locus by

$$\mathcal{B}_{\text{val}}(F) = \{x \in K : \text{there exists } \beta \in \Gamma \text{ with } v(F^n(x)) \geq \beta \text{ for all } n \geq 0\}.$$

Always $\mathcal{B}_{\text{int}}(F) \subseteq \mathcal{B}_{\text{val}}(F)$. The opposite inclusion is often silently assumed in a rank-one argument. Here it has an exact order-unit criterion.

For $x \neq 0$, set $\alpha(x) = \max(0, -v(x))$, and put $\alpha(0) = 0$. Thus α measures only how far the initial point lies outside the integral disk.

Theorem 10.1 (Exact bounded-orbit locus). *For $F = q^{-1}P$ as in (3.1),*

$$\mathcal{B}_{\text{val}}(F) = \mathcal{B}_{\text{int}}(F) \cup \{x \in K : \kappa + \alpha(x) \text{ is not an order unit in } \Gamma\}. \quad (10.1)$$

Consequently:

- (a) *If κ is an order unit, $\mathcal{B}_{\text{val}}(F) = \mathcal{B}_{\text{int}}(F)$.*
- (b) *If κ is not an order unit, then $\mathcal{B}_{\text{val}}(F) = \{x : \kappa + \alpha(x) \text{ is not an order unit}\}$. In this case the bounded-orbit locus is independent of P within the specified family, and it contains all of $\mathcal{O}$.*
- (c) *If Γ has no order unit, then $\mathcal{B}_{\text{val}}(F) = K$.*

Proof. If an orbit ever leaves $\mathcal{O}$, write its first negative valuation as $-a$, with $a > 0$. If the initial point was integral, then $0 < a \leq \kappa$: the preceding point was integral, P has integral coefficients, and $v(F(z)) \geq -\kappa$ for integral z . If the initial point was not integral, we may start at time zero with $a = \alpha(x)$.

Once negative, valuations obey (6.2). After a further n steps they are

$$-d^n a - (1 + d + \cdots + d^{n-1})\kappa. \quad (10.2)$$

For $n \geq 1$, the positive quantity whose negative occurs here lies between $d^{n-1}(a + \kappa)$ and $d^n(a + \kappa)$. Its values are cofinal in $\Gamma_{>0}$ exactly when $a + \kappa$ is an order unit. Indeed, if that element is an order unit, powers of d dominate any finite multiple of it; otherwise a single η exceeding all its integer multiples bounds the whole sequence. A finite initial orbit segment does not affect boundedness.

When the initial point is integral and later exits, $a + \kappa$ lies between κ and 2κ , so it is an order unit exactly when κ is. Otherwise $a + \kappa = \alpha(x) + \kappa$. Points which never leave are already in $\mathcal{B}_{\text{int}}(F)$. This proves (10.1). The three consequences follow directly; for (b), all points of $\mathcal{B}_{\text{int}}(F)$ have $\alpha = 0$, so the first set in the union is redundant. $\square$

10.1 A coarsened valuation-ring interpretation

Suppose Γ has an order unit u , and define

$$H = \{\gamma \in \Gamma : n|\gamma| < u \text{ for every integer } n \geq 1\}.$$

This is the maximal proper convex subgroup of Γ . In particular, a positive element belongs to H exactly when it is not an order unit. For clarity, the relevant elementary argument is included. Closure under addition follows by applying the condition to $2n|\gamma|$ and $2n|\delta|$: their sum is less than $2u$, hence $n(|\gamma| + |\delta|) < u$. Convexity is immediate. If $\gamma > 0$ is not an order unit, some η exceeds all $n\gamma$, while $\eta \leq Nu$ for some N . Applying the bound to $Nn\gamma$ gives $n\gamma < u$. Conversely that last condition prevents γ from being an order unit. Any convex subgroup containing an order unit is all of Γ , proving maximality. (Placement note: H is the kernel of the order-preserving homomorphism $\Gamma \rightarrow \mathbb{R}$ of `ent:lem:coarsening` in `entire-functions-at-arbitrary-rank`, with $h = u$. That report states the lemma under its standing divisibility hypothesis. The argument just given does not use divisibility.)

Corollary 10.2. *If Γ has an order unit but κ is not one, then*

$$\mathcal{B}_{\text{val}}(F) = \{x \in K : v(x) + H \geq 0 \text{ in } \Gamma/H\},$$

the valuation ring of the coarsened valuation. Relative to this coarsening, F has integral coefficients and unit leading coefficient.

Proof. Here $\kappa \in H$, and for $\alpha \geq 0$ the sum $\kappa + \alpha$ is not an order unit exactly when $\alpha \in H$. The formula follows from [theorem 10.1](#). Also $v(q) \in H$, so q is a unit for the coarsened valuation. The coefficients of P are integral and its leading coefficient is a unit already for the original valuation. $\square$

Example 10.3 (Escaping the disk without an unbounded orbit). With $\Gamma = \mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ and $\kappa = (0, 1)$, $H = \{0\} \oplus \mathbb{Q}$. Therefore $\mathcal{B}_{\text{val}}(F)$ consists of those elements whose valuation has nonnegative first coordinate. For $x = t^{(0, -1)}$ the integral disk has already been left, and (10.2) gives increasingly negative second coordinates. All valuations remain greater than $(-1, 0)$, so the orbit is still valuation-bounded. This is not a failure of the recurrence; it is the distinction between a direction that is not an order unit and a genuinely unbounded sequence in the full ordered group.

11 A general polynomial theorem when there is no order unit

The absence of an order unit has a consequence beyond the simple-reduction family. The following quantitative estimate explains why the native fine topology can conceal every finite-degree polynomial expansion.

Proposition 11.1 (An orbit and difference budget). *Let $f(X) = \sum_{j=0}^d a_j X^j \in K[X]$ have degree $d \geq 1$. Choose $A > 0$ with $v(a_j) \geq -A$ for all nonzero coefficients, and put $D_A = \{z : v(z) \geq -A\}$. For $x, y \in D_A$ and $n \geq 0$,*

$$v(f^n(x)) \geq -(d+1)^n A, \quad (11.1)$$

$$v(f^n(x) - f^n(y)) \geq v(x - y) - ((d+1)^n - 1)A. \quad (11.2)$$

Proof. Set $M_n = (d+1)^n A$. If $v(z) \geq -M_n$, every polynomial term has valuation at least $-A - dM_n \geq -(d+1)M_n = -M_{n+1}$. This proves (11.1) by induction.

For z, w with valuations at least $-M_n$, the divided difference

$$\frac{f(z) - f(w)}{z - w} = \sum_{j=1}^d a_j \sum_{\ell=0}^{j-1} z^\ell w^{j-1-\ell}$$

(where the right side also defines it when $z = w$) has valuation at least $-A - (d-1)M_n \geq -dM_n$. Apply this with $z = f^n(x)$ and $w = f^n(y)$. Accumulating the possible losses gives

$$d \sum_{j=0}^{n-1} M_j = ((d+1)^n - 1)A,$$

which is (11.2). Cancellation can only increase these valuations. $\square$

Theorem 11.2 (Universal native equicontinuity criterion). *The following statements are equivalent for $K = k((t^\Gamma))$:*

- (i) Γ has no order unit.
- (ii) Every orbit of every polynomial in $K[X]$ is valuation-bounded.
- (iii) For every polynomial f , its family of iterates is locally equicontinuous at every point of K , in the native valuation uniformity.

Under (i), the family is in fact uniformly equicontinuous on every D_A for which the coefficient bounds in proposition 11.1 hold.

Proof. Assume (i). For the A in proposition 11.1, choose $\eta > rA$ for every ordinary integer r . Then (11.1) bounds every orbit starting in D_A below by $-\eta$. Every individual starting point lies in some such D_A , proving (ii).

For any requested output threshold γ , choose $\delta = \max(\gamma, 0) + \eta$. If $x, y \in D_A$ and $v(x - y) > \delta$, then (11.2) gives $v(f^n(x) - f^n(y)) > \gamma$ for every n . This proves uniform equicontinuity on D_A . Such a disk is a neighborhood of each of its points: the ball $\{y : v(y - x) > -A\}$ is contained in it. Thus (iii) follows. Constant polynomials need no estimate and satisfy the same conclusions.

Conversely, suppose $u > 0$ is an order unit. The linear polynomial $f(X) = t^{-u}X$ has the orbit $f^n(1) = t^{-nu}$, whose valuations have no lower bound. Hence (ii) fails. Its iterates are not locally equicontinuous at zero: given any input threshold δ , choose N with $Nu > \delta$ and take $h = t^{Nu}$. Then h belongs to that input neighborhood, but $v(f^{N+1}(h)) = -u < 0$. Even the fixed output threshold 0 cannot be maintained. Thus (iii) fails as well. $\square$

Definition 11.3 (A deliberately named affine notion). For a polynomial f , define $J_{\text{aff},v}(f)$ to be the set of points of K having no native open neighborhood on which all iterates of f are equicontinuous as maps to the additive valued field K .

Corollary 11.4. *If Γ has no order unit, then $J_{\text{aff},v}(f) = \emptyset$ for every polynomial f .*

Proof. This is exactly statement (iii) of theorem 11.2. $\square$

The subscript is essential. Classical non-Archimedean Fatou and Julia sets are ordinarily defined on a projective line using a spherical metric, or on a Berkovich space, not by the specifically affine replacement; compare Definitions 4.15–4.16 of [4]. We claim neither that every Berkovich Julia set is empty nor that a rank-one Julia set disappears in its own topology. The corollary is a precise obstruction to one naive transplantation of the definition.

Example 11.5 (No uncountable-cofinality hypothesis is needed). Let $\Gamma = \bigoplus_{j \geq 0} \mathbb{Q}e_j$, ordered by the largest index having a nonzero coefficient, with its sign deciding the order. Then $e_{j+1} > ne_j$ for every positive integer n , and every group element has finite support. The group is countable and has no order unit. All conclusions of theorem 11.2 hold. This is a finite polynomial growth-budget phenomenon, not a consequence of assuming uncountable cofinality for the entire value group. (Placement note: here, as in `ent:rem:one-name`, cofinality means $\text{cf}(\Gamma)$, which is $\aleph_0$ for this group. As an ordered group it is the group $\Gamma_\infty = \bigoplus_{j \geq 0} \mathbb{Q}\omega^j$ of `ent:subsec:gamma-infinity`, via $e_j = \omega^j$.)

12 The surreal and surcomplex specialization

12.1 What is imported from surreal theory

Conway normal form identifies surreal numbers with set-supported well-based sums of real coefficients times monomials ω^a . The monomial law $\omega^{a+b} = \omega^a \omega^b$ identifies a set-sized ordered additive subgroup $\Gamma \subseteq \mathbf{No}$ with a Hahn monomial group by $t^\gamma = \omega^{-\gamma}$. Thus $\mathbb{R}((t^\Gamma))$ embeds into $\mathbf{No}$, and coefficientwise complexification embeds $\mathbb{C}((t^\Gamma))$ into $\mathbf{No}[i]$. This is classical normal-form theory, not a theorem first proved here; see Gonshor [2] and the explicit Hahn description in Berarducci–Mantova, Sections 2.3, 2.4, and 2.6 [3].

The valuation convention remains $v(\omega^{-\gamma}) = \gamma$. For a surcomplex number written in normal form, coefficients are in $\mathbb{C}$ and supports are still sets. Every finite list of parameters is contained in a common set-sized Hahn workspace obtained by taking the additive group generated by its supports and 1, so that the workspace value group is nonzero even for purely ordinary parameters. All infinite sums used to construct the centers have the common set-sized support certificate (4.4).

There is consequently no need to posit a set containing all surreals, or to sum over a proper class of exponents. Statements about an arbitrary additional point are proved after enlarging the workspace to include that point's support. The same center is obtained after every such enlargement by theorem 8.1.

12.2 An explicit quadratic theorem inside the actual field

Set

$$q = \omega^{-1}, \quad F(x) = \omega(x^2 - 1), \quad I = \{h : v(h) > n \text{ for every } n \in \mathbb{N}\}. \quad (12.1)$$

Here h may be surreal or surcomplex. In the real case, I consists of numbers smaller in magnitude than every positive ordinary power of ω^{-1} ; the valuation definition is the one used in the proofs. It contains $\omega^{-\omega}$.

Theorem 12.1 (A surcomplex full shift with infinitesimal fibers). *There is a canonical real surreal number*

$$x_s = \sum_{n \geq 0} a_n(s) \omega^{-n} \quad (s \in \{+1, -1\}^{\mathbb{N}})$$

for every binary itinerary. The locus whose every iterate is a finite surcomplex number is

$$\{x \in \mathbf{No}[i] : F^n(x) \text{ is finite for every } n\} = \prod_s (x_s + I). \quad (12.2)$$

Its restriction to $\mathbf{No}$ is obtained by using real $h \in I$. Every actual surcomplex preperiodic point of F is one of the real centers with eventually periodic itinerary. For $h = \omega^{-\omega}$,

$$v(F^n(x_s + h) - x_{\sigma^n s}) = \omega - n \quad (n \in \mathbb{N}). \quad (12.3)$$

In the full fine topology the fibers and their union are clopen. Every subset of the center set is closed and discrete. The itinerary map is continuous but not a quotient map to the product-topological full shift; its native quotient is discrete. The iterates are uniformly equicontinuous on the nonescaping locus. Every compact set-sized forward-invariant subset is finite and consists of preperiodic centers.

Proof. Apply the constant-coefficient construction in the workspace with exponent group $\mathbb{Z}$. This produces every x_s and its strong series. For any proposed point or perturbation, enlarge to a set-sized exponent group containing its support and 1. The exact-fiber theorem gives the same formula there, and the center is unchanged. Finite surreal and surcomplex elements are exactly the elements of nonnegative natural valuation, so the left side of (12.2) is precisely the integral nonescaping condition.

The class I is nonzero because $\omega > n$ for all n . The fiber formula and (12.3) follow from [theorem 5.1](#). All coefficients of the binary centers are real. Any preperiodic surcomplex point is a root of a polynomial which already splits into distinct real linear factors by [theorem 6.1](#), so it is one of those real centers. For topology, the proofs in [theorems 7.1](#) and [7.5](#) only use balls, a fixed dominating valuation, and set-indexed compactness; they apply verbatim. $\square$

Remark 12.2 (Fine topology versus a workspace topology). The natural valuation topology of the full surreal field is its order (fine) topology: every positive radius exceeds some monomial radius, and increasing a monomial valuation gives a radius strictly smaller than any prescribed positive element. In the surcomplex field the same statement holds with modulus balls, or equivalently with the product of the two real-coordinate topologies. These assertions also follow directly by comparing a leading monomial with its nonzero real or complex coefficient.

The restriction of this full topology to $\mathbb{R}((\omega^{-1}))$ is not that field's intrinsic ω^{-1} -adic topology. In the latter, the center set is a Cantor set. In the former, the same set is uniformly discrete, since a radius with valuation ω separates every pair of centers. These are two different topologies on the same algebraic objects, not two incompatible conclusions.

Remark 12.3 (Placement note: the set-level clauses are special cases). The manuscript did not cite *Surcomplex Analysis* (docs/surcomplex/analysis). That report's fine topology (a: def: fine) is the full topology used in [theorem 12.1](#) and [remark 12.2](#). Its a: thm: discrete proves that every set $A \subseteq \mathbf{No}[i]$ is closed and discrete in that topology, and that one radius separates all pairs of distinct points of A . The following consequences use nothing from this report except, in the second item, the identification of the preperiodic points.

- Every subset of the center set $\{x_s\}$ is closed and discrete, because that set is indexed by the set $\{+1, -1\}^{\mathbb{N}}$.
- Every compact set-sized subset is finite. In particular, every compact forward-invariant set-sized subset of the locus (12.2) is finite. Its points have finite orbits, so they are preperiodic, and by theorem 12.1 they are real centers.
- The uniform separating radius of `a:thm:discrete(ii)` is realized explicitly by the radius of valuation ω in remark 12.2. The comparison made in that remark is an instance of `a:rem:notvaluation`: the fine topology induced on a set-sized Hahn subfield is strictly finer than the subfield's own valuation topology.

The set-level discreteness and compactness clauses in the last sentence of theorem 12.1 are therefore special cases of `a:thm:discrete`. The substance of theorem 7.1(b) and theorem 7.5 lies in a set-sized workspace $k((t^\Gamma))$ with its intrinsic valuation topology. That topology is not discrete, and in the order-unit case the center set is a Cantor set (theorem 7.1(a)). Some statements concern the fibers $x_s + I$, which are proper classes, or whole fine balls. These are the openness of the fibers, the failure of π to be a quotient map, uniform equicontinuity on the locus (12.2), and corollary 12.4. None of them is a consequence of `a:thm:discrete`.

12.3 Exact coefficients and the first visible digits

For the quadratic family the inverse branches are

$$\Phi_\varepsilon(w) = \varepsilon \sqrt{1 + qw} = \varepsilon \sum_{r \geq 0} \binom{1/2}{r} q^r w^r, \quad \varepsilon \in \{+1, -1\}.$$

The root is the formal root with constant coefficient 1. Its Hahn evaluation is justified by the positive valuation of qw . Write $a_n(s)$ for the coefficient of q^n in x_s . Comparing coefficients in $x_s^2 - 1 = qx_{s\sigma}$ gives

$$a_0(s) = s_0, \quad a_n(s) = \frac{a_{n-1}(\sigma s) - \sum_{j=1}^{n-1} a_j(s) a_{n-j}(s)}{2s_0} \quad (n \geq 1). \quad (12.4)$$

In particular,

$$x_s = s_0 + \frac{s_0 s_1}{2} q + \left(\frac{s_0 s_1 s_2}{4} - \frac{s_0}{8} \right) q^2 + \cdots. \quad (12.5)$$

All coefficients are rational. The coefficient at order n sees exactly a finite prefix, although the resulting Hahn point carries an arbitrary infinite word.

For the constant positive itinerary, the fixed point is

$$\begin{aligned} x_+ &= \frac{q}{2} + \sqrt{1 + \frac{q^2}{4}} \\ &= 1 + \frac{q}{2} + \frac{q^2}{8} - \frac{q^4}{128} + \frac{q^6}{1024} - \frac{5q^8}{32768} + \cdots. \end{aligned} \quad (12.6)$$

The other fixed point is $x_- = q/2 - \sqrt{1 + q^2/4}$. The point $x_+ + \omega^{-\omega}$ is not periodic, but every one of its iterates has residue +1. Formula (12.3) records its exact deviation from the fixed point at every finite time.

A deformation gives another explicit check. Replace $P = X^2 - 1$ by $\tilde{P} = X^2 - 1 + \varepsilon$, with $\varepsilon = \omega^{-\omega}$. Then

$$\tilde{x}_+ = \frac{q}{2} + \sqrt{1 - \varepsilon + \frac{q^2}{4}}, \quad [\omega^{-\omega}](\tilde{x}_+ - x_+) = -\frac{1}{2}.$$

The bracket denotes the normal-form coefficient of that monomial. The two maps have exactly the same labeled nonescaping fibers, but no common fixed point. This is the simplest instance of [corollary 9.3](#).

12.4 Polynomial iteration in the full fine topology

For every positive surreal A there is a surreal η exceeding all ordinary integer multiples of A , for example $\eta = \omega A$. The quantitative estimates in [proposition 11.1](#) therefore apply to any fixed polynomial with surreal or surcomplex coefficients and any set of inputs in the specified ball. All relevant parameters and η can first be placed in a common set-sized workspace.

Corollary 12.4. *Every orbit of every finite-degree polynomial over $\mathbf{No}$ or $\mathbf{No}[i]$ is bounded by some surreal modulus bound. The family of iterates of each such polynomial is locally equicontinuous everywhere in the full fine uniformity. Its explicitly affine equicontinuity obstruction $J_{\text{aff},v}$ is empty.*

Proof. For a fixed polynomial and an initial ball, choose the bound A in [proposition 11.1](#) and then a larger scale η dominating all nA . The proof of [theorem 11.2](#) uses only this particular pair of scales. For each pair of inputs, enlarge the set-sized workspace to contain their supports as well. The same A , η , and input threshold work in every such enlargement, so the estimate is uniform on the entire fine ball, even though that ball is not itself set-sized. A lower valuation bound gives a modulus bound by taking a still larger monomial to absorb every possible finite leading coefficient. The same comparison of monomial balls and modulus balls transfers equicontinuity to the fine uniformity. $\square$

For the explicit map $F(x) = \omega(x^2 - 1)$, the orbit of 0 satisfies

$$v(F^n(0)) = -(2^n - 1) \quad (n \geq 1).$$

It leaves the integral disk immediately, yet every iterate has modulus less than ω^ω . Thus $0 \in \mathcal{B}_{\text{val}}(F) \setminus \mathcal{B}_{\text{int}}(F)$ in the full surreal field. More simply, the ordinary sequence 2^{2^n} is bounded by an infinite surreal number although it is unbounded over $\mathbb{R}$. The content needed for polynomial equicontinuity is stronger than mere existence of a bound for one sequence: (11.2) provides a single input precision that works for every iterate and every pair of inputs in a whole prescribed ball. The corollary still does not assert that classical or Berkovich Julia sets are empty in their own topologies.

13 Verification, reproducibility, and formalization

The accompanying program code/`verify.py` uses Python's standard library only, with exact rational coefficients and integer exponent pairs. It is a finite consistency test of the displayed algebra, not a substitute for the proofs of infinite support or compactness.

The recorded run performed **38,139 exact checks**, all passing. The principal counts are shown in [table 1](#). Every count is computed by the shipped program; the full record is in `data/verification.json`.

13.1 The two coefficient algorithms

The first algorithm composes the inverse branches $\varepsilon\sqrt{1+q\omega}$ using a truncated binomial evaluator. The second uses (12.4) directly, constructing the tail coefficients before the coefficients of

Table 1: Exact finite checks in the included run.

Check	Number
Two independent center constructions, all binary words of length 8	256
Itinerary equation through the available truncation order	256
Displayed first three coefficients	256
Every unordered pair of length-8 words: exact first mismatch order	32,640
Periodic identities, all binary words of periods 1 through 6	126
Möbius exact-period counts	6
Two-parameter identity, ideal divisibility, and first response	96
Displayed positive fixed-point expansion	1
Rank-two error valuation and leading coefficient, times 0 through 10	22
Exterior valuation recurrence versus its closed formula	4,480
Total	38,139

the current word. These are different algorithms, though both implement the same defining polynomial equation; their agreement is not independent foundational verification.

For a requested coefficient order N , only $N + 1$ symbols are needed. The direct recursive implementation has polynomial arithmetic cost in N ; with straightforward convolutions and all required suffixes, $O(N^3)$ rational arithmetic operations suffice. Coefficient bit growth is not included in this arithmetic-operation bound. Nothing in this finite computation decides membership in I_κ , which quantifies over all finite q -orders.

The bivariate checks work in $\mathbb{Q}[q, \varepsilon]$ truncated by total degree. They verify $x_s^2 - 1 + \varepsilon = qx_{os}$, specialization at $\varepsilon = 0$, and the coefficient $[q^0 \varepsilon^1](\tilde{x}_s - x_s) = -s_0/2$. The rank-two checks track the formal difference recurrence

$$h_{n+1} = q^{-1}(2x_+h_n + h_n^2)$$

in powers of an additional symbol h , with lexicographically ordered exponents $(\deg_h, \deg_q)$. Their leading terms are $2^n h q^{-n}$, of valuation $(1, -n)$. Discarded higher h degrees cannot affect the degree-1 leading term. These checks do not pretend to turn an approximate fixed-point truncation into an exact Hahn fixed point; the exact point was proved in (12.6).

13.2 How to reproduce the package

The manuscript's instructions, for its delivered layout, were: from the package directory, run

```
python code/verify.py --output data/verification-rerun.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The source has an internal bibliography and needs no BibTeX run. `make all` performs the verification and PDF build. Python 3.10 or later is sufficient, with no network access or external Python packages required. Run without `-O`. Elapsed time and Python version in the JSON are environment-dependent; the exact check counts and rational outputs are deterministic.

Placement notes. The program and its record are kept byte-identical to the delivered files. Three facts about the placed layout matter.

- (i) The program's only option is `--output`. Its default is `data/verification.json` in the directory that contains `code/`, whatever the current directory. That file is the shipped

record, and the program writes its output there without checking whether the file exists. The command in its docstring names the same path. A bare `python code/verify.py` therefore overwrites the recorded evidence. The elapsed-time field changes on every run, and the Python-version field changes on another interpreter. Run the program on a copy of this directory, or pass an explicit `--output` path outside it.

- (ii) The delivered Makefile is placed at `code/Makefile`. Its recipes use paths relative to the report directory, so it is invoked from there as `make -f code/Makefile verify`. That target writes the new file `data/verification-rerun.json` and leaves the record untouched.
- (iii) The collection builds the article with `latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex`.

At placement the program was rerun with Python 3.14.4, writing to a scratch path. It reported PASS with 38,139 checks. Its output agreed with `data/verification.json` in every field except the Python version and the elapsed time.

13.3 Current Lean coverage and the remaining route

A useful formalization boundary separates three layers. The first is finite algebra: the divided-difference identity, simple-residue uniqueness, exact valuation changes, and the orbit budget. The second constructs the universal formal centers and evaluates them using the finite-parameter Hahn map. The third proves the fiber decomposition, the cylinder equivalence, and the topological dichotomy.

Once a general center-construction interface supplies $F(x_s) = x_{\sigma s}$ and the correct residues, the exact-fiber, preperiodicity, and compact-subsystem proofs require no further infinite summation. This isolates the support-heavy part of the remaining Lean development.

The current mapping in `docs/FORMALIZATION.md` records [definition 2.1](#) and [lemma 2.3](#) in `Surreal/HahnSeries/ScaleIdeal.lean`, over an arbitrary coefficient field and ordered abelian group. It proves the convex scale subgroup, the error ideal and its q -invariance, the intersection formula, the order-unit criterion, and the coarsened valuation-ring description. Clopenness is expressed by explicit valuation-ball conditions; no Mathlib topology on Hahn series is presumed. The coarsening is represented by a valuation subring, without an explicit ordered quotient isomorphism.

The positive-support lemma is also proved in the collection's `Surreal/HahnSeries/Neumann.lean`, including `neumann_positive` and `finite_words_of_sum_eq`. The universal centers, exact fibers and later dynamical theorems are not formalized by these modules. The report's written proofs and finite regression checks are separate from this limited Lean coverage.

14 Limitations and further research questions

The hypotheses in (3.1) are doing real work. Simple residue roots make the divided difference a unit, provide unique branches, and make every derivative along the integral orbit have the same valuation loss. A repeated residue root can require fractional scales, ramification, or nonunique inverse branches. None of the present theorems silently covers that situation.

The polynomial has finitely many coefficients and finitely many simple branches. The finite-parameter evaluation proof does not authorize an arbitrary countable family of independent

perturbation parameters without a new joint support hypothesis. Likewise, the general orbit budget is for polynomials. Rational maps can repeatedly approach poles; one cannot apply the polynomial bound to their denominators without controlling those approaches.

Question 14.1 (Nonuniform branch scales). For a polynomial or rational map with finitely many inverse branches whose exact gains are different positive values κ_i , can one construct canonical itinerary centers and prove that the fiber of s is precisely the set of errors whose valuations exceed all partial sums $\sum_{j < N} \kappa_{s_j}$? What finite-parameter support conditions are sufficient when the branch equations are not simultaneously normalized by one q ?

The metric half of this proposed statement would follow from the same pullback calculation. The nontrivial remaining issue is realization of every admissible itinerary with a uniform support certificate. The present article does not assume this generalization.

Question 14.2 (A topology retaining both levels of information). Can one give a useful enriched dynamical space that retains the compact full-shift quotient and the nontrivial native infinitesimal fibers simultaneously, without declaring the quotient topology to be the wrong one? A precise comparison with higher-rank adic or other valued geometric spaces would be preferable to relabeling the quotient as a Berkovich space without a construction.

Question 14.3 (Dynamics inside a single fiber). For a periodic itinerary, what canonical normal forms classify the restriction of the return map to $x_s + I_\kappa$? The valuation law is exact, but it records only the size of a deviation, not its coefficients at all finer scales. Which part of that finer structure survives the flat deformations of [corollary 9.3](#)?

These are proposed continuation problems arising from the proved framework, not assertions that the same questions are established open problems in the published literature.

15 Conclusion

The infinite word in an inverse-branch construction determines a point only when the accumulated scales $n\kappa$ are cofinal in Γ , that is, when κ is an order unit. Otherwise it determines an exact affine infinitesimal fiber. Strong Hahn evaluation supplies the canonical representative that topological convergence cannot supply, and the exact divided-difference metric identifies every other representative.

For the polynomial family studied here, this produces a sharp and fully explicit separation between finite algebraic orbit data and infinite native topological behavior. All preperiodicity equations split in the original workspace. Enlarging the exponent group can leave those equations and their roots unchanged while thickening every itinerary fiber, destroying compact infinite invariant subsystems, and turning the same center set from a Cantor space into a discrete space. Infinitesimal coefficient changes beyond every power of the dynamical scale preserve the symbolic partition but move the actual fixed points.

The surreal example $F(x) = \omega(x^2 - 1)$ is therefore not merely an ordinary real or complex dynamical system with a larger coefficient field. Its behavior depends on which scales the chosen topology can resolve. The correct conclusion is neither that ordinary chaos survives verbatim nor that all dynamics becomes trivial: the symbolic quotient, the exact fibers, the algebraic periodic locus, and the ambient topology retain different, explicitly calculable information.

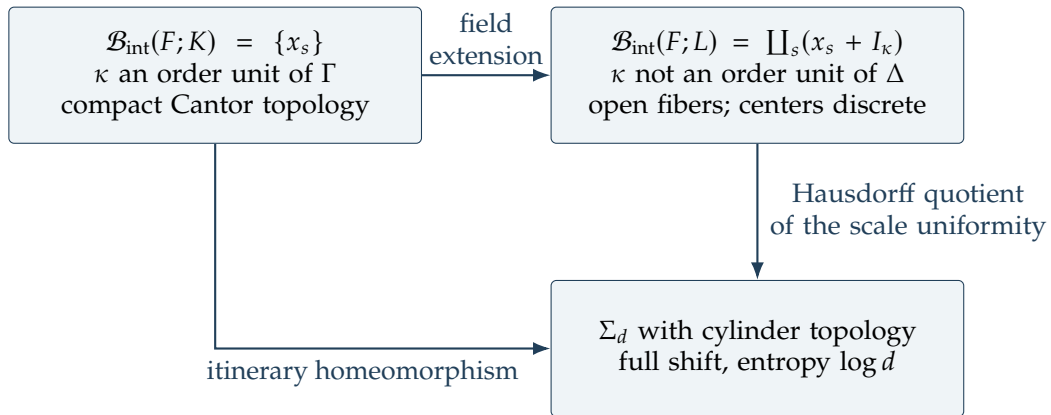
Table 2: Logical dependencies and status of the principal results.

Result	Dependencies and status
Positive-support lemma	Classical imported result; cited to Neumann. (Placement: proved in the collection’s Lean library, section 13.3 .)
Finite-parameter evaluation	Explicit consequence with coefficient-finiteness proof in lemma 2.5 .
Universal centers	Formal implicit recursion, formal Q -adic completeness, and strong evaluation; proved in theorem 4.1 .
Exact fibers	Centers plus the exact divided-difference metric; proposed new arbitrary-rank formulation, theorem 5.1 .
Preperiodicity splitting	Exact fibers and finite polynomial degree; proved in theorem 6.1 . Counting formulas are classical consequences.
Native topology and compact rigidity	Exact first mismatch, open error ideal, and elementary compactness; proposed new package, theorems 7.1 and 7.5 . (Placement: the proposal concerns set-sized workspaces; in No[i] the set-level clauses are special cases of <code>a:thm:discrete</code> , remark 12.3 .)
Scalar extension and transcendence	Universal support, preperiodicity splitting, and one-term valuation dominance; theorem 8.1 .
Deformation and flat invisibility	Formal perturbation ideal before evaluation; theorem 9.1 and corollary 9.3 .
Bounded orbit locus	Exact exterior valuation recurrence and order-unit criterion; theorem 10.1 .
General polynomial equicontinuity	Finite-degree growth budget; structural consequence of no order unit, theorem 11.2 .
Surreal realization	Classical Conway normal form plus the set-sized workspace proofs; theorem 12.1 and corollary 12.4 .

A Dependency and claim ledger

All newly proved assertions are mathematical claims of this draft. No independent peer review is claimed. The precise partial Lean coverage is stated in [section 13.3](#); it does not certify the remaining dynamical theorems. Finite tests support only the finite identities and regression cases listed in [section 13](#); they do not establish the general statements in this ledger.

B A compact diagram of the topology change



The upper arrow is an algebraic inclusion of point sets; it is not claimed continuous for the two intrinsic native topologies. The right-hand arrow is continuous in the native topology

but is not its quotient map. The label on that arrow specifies the coarser uniformity whose Hausdorff quotient actually is the full shift.

C Limitations and non-claims, collected

This appendix was added at placement. It gathers the limitations, disclaimers, and non-claims that the manuscript states in several places (its article, its delivered README, the delivered RESEARCH_STATUS.md, and the limitations field of data/verification.json). Items S1–S28 come from the source and were regrouped here. Their historical no-formalization wording is updated to the current coverage in [section 13.3](#). Items P1–P3 were added at placement. Nothing in the body is weakened by this list, and nothing outside it is claimed.

Status, priority, and provenance

- S1. The report is an unrefereed research draft. Its partial machine-checked coverage is specified in [section 13.3](#); the remaining dynamical claims have no such certificate.
- S2. Priority is not certified. The fiber, compact-subsystem, and extension/deformation package is *proposed* as new after a targeted literature search whose results were incomplete and sometimes noisy. Failure to locate the statements is not proof that they have never appeared, and the claims need independent expert review before publication as new ([section 1](#); RESEARCH_STATUS.md).
- S3. No named published open conjecture is claimed solved. The three questions of [section 14](#) are proposed continuations, not established open problems.
- S4. Classical inputs are credited, not claimed. These are the positive-support lemma, formal implicit-function recursion, valuation coarsening, the one-sided full shift and its periodic counts and zeta series, the rank-one Cantor construction, and Conway normal form ([sections 1, 6 and 12](#)).
- S5. The delivered manuscript included no Lean file and ran no repository build. The current collection does formalize the scale subgroup and error-ideal lemma, as recorded in [section 13.3](#); this is not a formalization of the entire article.
- S6. The repository audit was targeted: the inventories and the dynamics report’s description only, with no line-by-line review. No assertion of complete absence of overlap across every file is made. During that original survey the repository was neither rebuilt nor modified; its descriptions are provenance, not validation ([section 1.1](#); RESEARCH_STATUS.md; bibliography).
- S7. The repository’s Hahn support machinery and its distinctions among analytic categories are not presented as discoveries ([section 1.1](#)).
- S8. The article does not purport to resolve the repository’s unrelated birthday, automorphism, or differential-equation questions.

Hypotheses

- S9. The theorems need a degree-preserving reduction whose roots are distinct and lie in k ; for $k = \mathbb{R}$ they must be real. Repeated residue roots, with fractional scales, ramification, or nonunique branches, are not covered ([section 14](#)).

- S10. Only finitely many coefficients and branches are allowed. An arbitrary countable family of independent perturbation parameters is not covered without a new joint support hypothesis.
- S11. The orbit budget is for polynomials. Rational maps, which can repeatedly approach poles, are not covered.
- S12. The nonuniform-branch-scale generalization of [question 14.1](#) is not assumed anywhere.

What the conclusions mean

- S13. *Valuation-expanding* does not mean repelling for a real-valued absolute value. The expanding points are not identified with classical Julia points ([section 1](#), [remark 7.2](#)).
- S14. Integral nonescape and valuation-boundedness are different properties ([section 10](#)).
- S15. Strong Hahn summation is an algebraic support operation, not a topological limit. Partial sums need not converge in the native topology, and the nested approximants do not converge when κ is not an order unit ([remarks 2.6, 4.3 and 9.2](#)).
- S16. The centers are canonical only relative to the fixed embedding $k \hookrightarrow K$, the reduction P_0 , and the universal construction. No intrinsic uniqueness among set-theoretic sections of π is asserted ([section 4](#)).
- S17. In [theorem 9.1](#), no equality is asserted when the leading numerator $D_\theta(c_{s_0})$ vanishes.
- S18. [Theorem 7.5](#) assumes $F(A) \subseteq A$, not $F(A) = A$, and does not assert that every point is periodic. There is no conflict with entropy monotonicity for compact factors, because the native space is noncompact.
- S19. The full shift is the Hausdorff quotient of the coarsened κ -scale uniformity, not the native quotient ([proposition 7.4](#)). In [Appendix B](#) the extension arrow is not claimed continuous, and the right-hand arrow is continuous but not a quotient map.
- S20. $J_{\text{aff},v} = \emptyset$ is a statement about the stated affine native notion only. It is not claimed that Berkovich or spherical Julia sets are empty, or that a rank-one Julia set disappears in its own topology ([corollaries 11.4 and 12.4](#)).
- S21. No set of all surreal numbers is formed and no proper-class sum is taken. Compactness in the surreal field refers to set-sized subsets. The fine and workspace topologies are two topologies, not two incompatible conclusions ([section 12](#)).
- S22. Primes are formal X -derivatives with the coefficients held fixed. No derivation on surreal coefficients is used, and no differential result of Berarducci–Mantova is used ([section 3](#), [\[3\]](#)).

Computations

- S23. The finite checks prove no infinite statement. They do not prove infinite Hahn summability. The transfinite-support, topology, compactness, and proper-class results are proved in the text, not checked by the program. Nothing computed decides membership in I_κ ([section 13](#); [data/verification.json](#)).
- S24. The two center algorithms are different but share the defining equation. Their agreement is not independent foundational verification.

- S25.** The $O(N^3)$ operation count excludes coefficient bit growth.
- S26.** The rank-two checks do not turn an approximate fixed-point truncation into an exact Hahn fixed point.
- S27.** The delivered finite verification uses no general surreal canonicalizer, floating-point simulation, or proof-assistant certificate (`data/verification.json`).
- S28.** Packaging (delivered README): the delivered PDF was not an accessible tagged PDF, and no third-party books, research-paper PDFs, or proof-assistant binaries were included. The PDF in this directory is rebuilt from the source and is not tagged either.

Added at placement

- P1.** No result is transferred into or out of this report from `dynamics-and-normal-forms`, `analysis`, `entire-functions-at-arbitrary-rank`, `hahn-tate-uniformization`, or `rank-one-berkovich`. This report answers neither the Julia/Fatou exclusion nor the repelling-multiplier exclusion of `dynamics-and-normal-forms` ([section 1.4](#)).
- P2.** For the class `No[i]` with its fine topology, the set-level discreteness and compactness clauses are special cases of `a:thm:discrete`. The proposal of novelty for those clauses concerns set-sized workspaces ([remark 12.3](#)).
- P3.** The order-unit dichotomy of [theorem 7.1](#) is not the cofinality dichotomy `ent:cor:cofinality`. The convex subgroup H_κ shares its definition with H_α of `hahn-tate-uniformization`. That identifies a subgroup, not a theorem.

References

- [1] B. H. Neumann, *On ordered division rings*, Transactions of the American Mathematical Society **66** (1949), 202–252. doi:10.1090/S0002-9947-1949-0032593-5. Used for the positive-support lemma for ordered series.
- [2] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series, vol. 110, Cambridge University Press, 1986. [Publisher record](#). Classical background for Conway normal form and surreal arithmetic.
- [3] A. Berarducci and V. Mantova, *Surreal numbers, derivations and transseries*, Journal of the European Mathematical Society **20** (2018), 339–390. [arXiv:1503.00315](#). The freely available version explicitly describes the Hahn normal-form structure; no differential result from that paper is used here.
- [4] R. L. Benedetto, *Non-Archimedean dynamics in dimension one: lecture notes*, Arizona Winter School, 2010, version of 9 March 2010. [Freely available lecture notes](#). Definitions 4.15–4.16 and Example 4.39 provide the classical topological and quadratic comparison, not the arbitrary-rank conclusions proved here.
- [5] V. Reshetnikov, *Surreal*, repository root and documentation inventories, snapshot `a3124af79f66b8b9c196d76b4cbc5ac3938907c4`, consulted 22 September 2026. [Pinned repository](#); [documentation inventory](#). Repository descriptions are provenance, not independent validation of this article.

- [6] *Surcomplex Dynamics: Linearization Thresholds, Periods, and Normal Forms*, README for docs/surcomplex/dynamics-and-normal-forms, in the same pinned repository [5]. [Pinned report description](#). Consulted for scope and overlap; the full merged manuscript was not exhaustively audited.